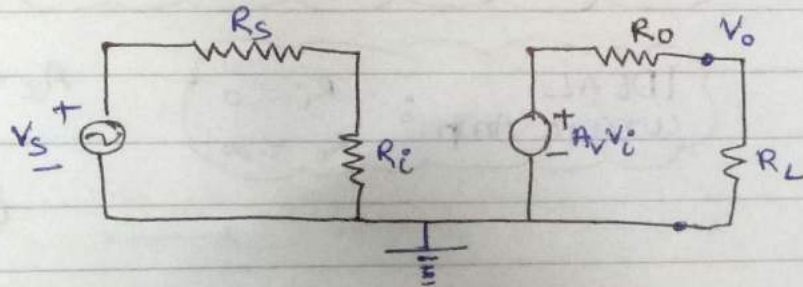


FEEDBACK IN AMPLIFIER

- i) voltage Amplifier
- ii) Current Amplifier
- iii) Trans conductance Amplifier
- iv) Trans resistance Amplifier

i) Voltage Amplifier



* Acha voltage amp vo koto hai jo jiska voltage gain stable ho * [It is not necessary that voltage gain is to be high]

→ It is said to be voltage amp. only if.

$$R_i \gg R_s \quad R_o \ll R_L$$

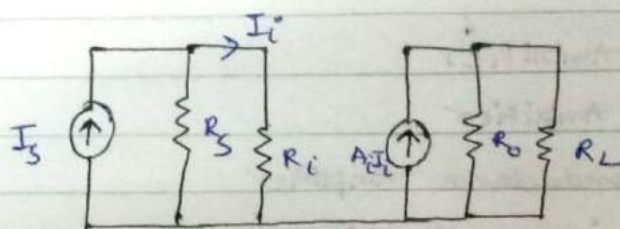
IDEAL VOLTAGE	:	$R_i \approx \infty$
AMPLIFIER	:	$R_o = 0$

$$A_v = \frac{V_o}{V_i} = A_{v0} \times \frac{R_L}{R_o + R_L}$$

IF $R_L \rightarrow \infty$

$$A_v = A_{v0}$$

ii) Current Amplifier



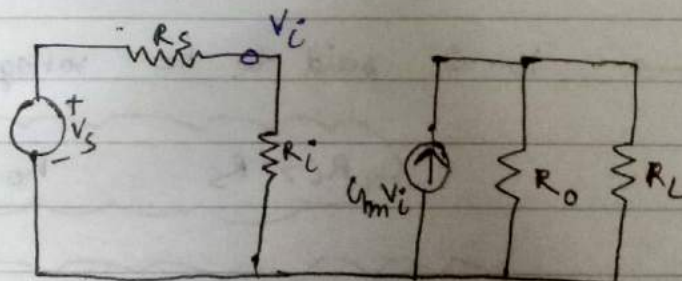
Condition : $R_i \ll R_s$ $R_o \gg R_L$

IDEAL
Current Amp : $R_i = 0$
 $R_o \approx \infty$

$$A_I = A_i \times \frac{R_o}{R_o + R_L}$$

$R_L \rightarrow 0$
 $A_I = A_i$

iii) Trans Conductance Amplifier



For stability

$R_i \gg R_s$
 $R_o \gg R_L$

IDEALLY

$R_i \approx \infty$
 $R_o \approx \infty$

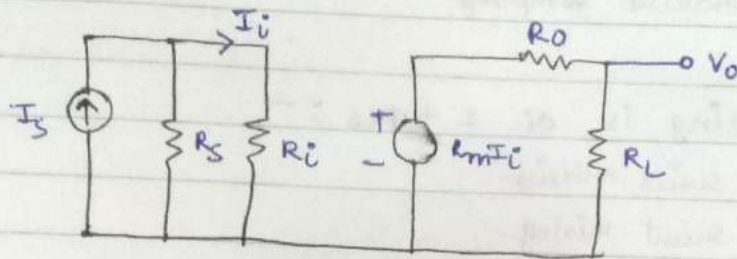
$$G_M = \frac{I_L}{V_i} = G_m \times \frac{R_o}{R_o + R_L}$$

$R_L \rightarrow 0$

$G_M = G_m$

→ Trans Conductance gain

iv) Trans Resistance Amplifier



For stability

$$R_i \ll R_s$$

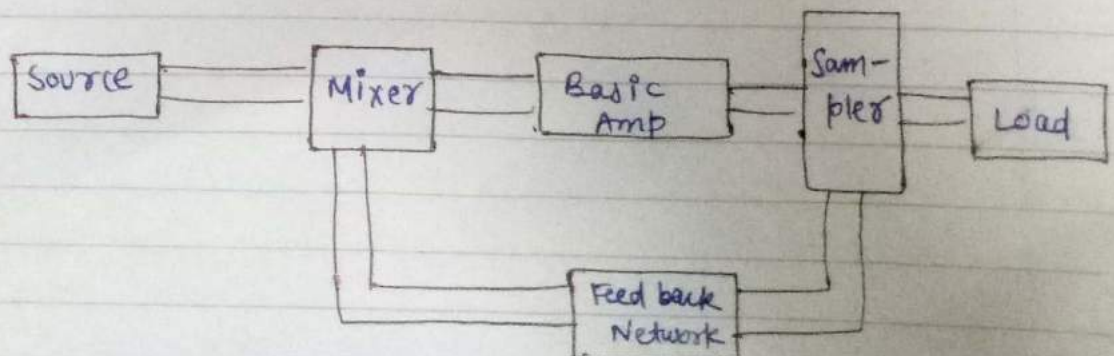
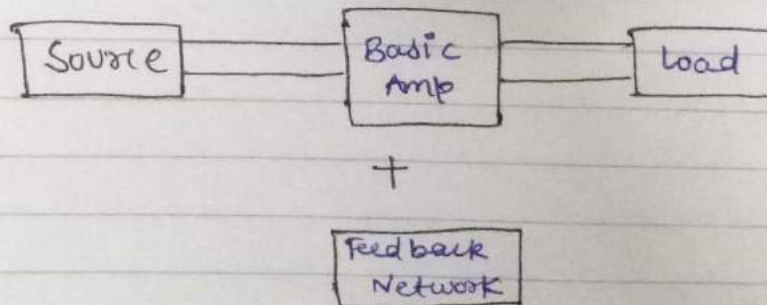
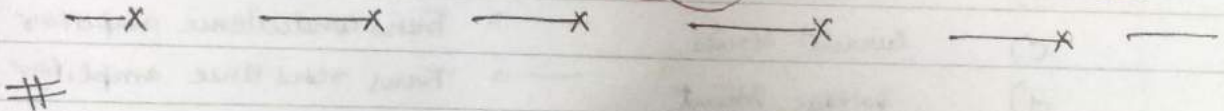
$$R_o \ll R_L$$

IDEAL

$$R_i = 0$$

$$R_o = 0$$

$R_m \rightarrow$ Trans Resistance
gain



$\rightarrow$ To improve the amp; we use feedback network.

∴ Sampling is of 2 types : -

- voltage sampling
- current sampling

∴ Mixing is of 2 types : -

- series mixing
- shunt mixing

Now using above combinatⁿ we have total case of Feedback is 4.

- | | | |
|-------------------|---|-----------------------------|
| a) Voltage series | → | Voltage amplifier |
| b) Current shunt | → | Current amplifier |
| c) Current series | → | Trans conductance amplifier |
| d) Voltage shunt | → | Trans resistance amplifier |